AP Assignment Report

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COMP1013 Analytics Programming

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Question 1: Data Inspection, Cleaning and Visualisation

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Rationale

The four data sets are imported independently due to the different types of data. Access to World Cup information. Relative project paths are used to ensure that the analysis can be run after downloading repository to another computer. Question Just like a number, a missing value is represented as 'NA', which is R's default representation for. missing data. The factors is applied to the following variables:The following variables are converted to factors: HomePenalty is not a continuous measurement, they are categories. 'AwayPenalty' are converted to numeric variables so that they can be analysed Correctly, any missing data are filled with zeros as per task requirements. The penalty shootout matches are chosen using 'subset()' and then plotted. A The choice of bar chart is chosen because 'HomePenalty' is a count variable that takes only discrete values and has a Few Integer values. exact frequency of each bar in separate figure Penalties are far more easy to identify with each interval in the grouped histogram than with each interval in the drawn histogram.

Load the required package

```
library(ggplot2)
library(here)
```

Load the datasets

```
# Load the four FIFA World Cup dataset from the project data folder.

 #Exercise 1
  Tournaments <- read.csv("E:/A2 data/Tournaments.csv")
  Teams <- read.csv("E:/A2 data/Teams.csv")
  Stadiums <- read.csv("E:/A2 data/Stadiums.csv")
  Matches <- read.csv("E:/A2 data/Matches.csv")
```

Inspect the data structure

```
# Inspect the dimensions, variable names, and original data types.

str(Tournaments)
```

```
## 'data.frame':    23 obs. of  2 variables:
##  $ TournamentID  : chr  "WC-1930" "WC-1934" "WC-1938" "WC-1950" ...
##  $ TournamentName: chr  "1930 FIFA World Cup" "1934 FIFA World Cup" "1938 FIFA World Cup" "1950 FIFA
```

```
str(Teams)
```

```
## 'data.frame':    85 obs. of  3 variables:
##  $ TeamID  : chr  "T-01" "T-02" "T-03" "T-04" ...
##  $ TeamName: chr  "Algeria" "Angola" "Argentina" "Australia" ...
##  $ TeamCode: chr  "DZA" "AGO" "ARG" "AUS" ...
```

```r
str(Stadiums)
```

```
## 'data.frame':    193 obs. of  4 variables:
##  $ StadiumID  : chr  "S-089" "S-108" "S-107" "S-111" ...
##  $ StadiumName: chr  "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ..
##  $ City       : chr  "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
##  $ Country    : chr  "Japan" "Qatar" "Qatar" "Qatar" ...
```

```r
str(Matches)
```

```
## 'data.frame':    964 obs. of  16 variables:
##  $ MatchID        : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID   : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName      : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v
##  $ Stage          : chr  "group stage" "group stage" "group stage" "group stage" ...
##  $ MatchDate      : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime      : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID      : chr  "S-193" "S-192" "S-192" "S-193" ...
##  $ HomeTeamID     : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID     : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore  : int  4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore  : int  1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime      : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ PenaltyShootout: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty    : chr  "?" "0" "0" "0" ...
##  $ AwayPenalty    : chr  "0" "0" "0" "?" ...
##  $ Result         : chr  "home team win" "home team win" "home team win" "home team win" ...
```

Present the data

```r
# Display the first six observations from each dataset.

head(Tournaments)
```

```
##   TournamentID      TournamentName
## 1      WC-1930 1930 FIFA World Cup
## 2      WC-1934 1934 FIFA World Cup
## 3      WC-1938 1938 FIFA World Cup
## 4      WC-1950 1950 FIFA World Cup
## 5      WC-1954 1954 FIFA World Cup
## 6      WC-1958 1958 FIFA World Cup
```

```r
head(Teams)
```

```
##   TeamID  TeamName TeamCode
## 1   T-01   Algeria      DZA
## 2   T-02    Angola      AGO
## 3   T-03 Argentina      ARG
## 4   T-04 Australia      AUS
## 5   T-05   Austria      AUT
## 6   T-06   Belgium      BEL
```

```r
head(Stadiums)
```

```
##   StadiumID           StadiumName      City Country
## 1     S-089          Jita Stadium      Jita   Japan
## 2     S-108 Ahmad bin Ali Stadium Al Rayyan   Qatar
## 3     S-107       Al Bayt Stadium   Al Khor   Qatar
## 4     S-111     Al Janoub Stadium Al Wakrah   Qatar
## 5     S-112    Al Thumama Stadium      Doha   Qatar
## 6     S-064         Allianz Arena    Munich Germany
```

```r
head(Matches)
```

```
##     MatchID TournamentID               MatchName       Stage MatchDate
## 1 M-1930-01      WC-1930         France v Mexico group stage 7/13/1930
## 2 M-1930-02      WC-1930 United States v Belgium group stage 7/13/1930
## 3 M-1930-03      WC-1930     Yugoslavia v Brazil group stage 7/14/1930
## 4 M-1930-04      WC-1930          Romania v Peru group stage 7/14/1930
## 5 M-1930-05      WC-1930      Argentina v France group stage 7/15/1930
## 6 M-1930-06      WC-1930          Chile v Mexico group stage 7/16/1930
##   MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore AwayTeamScore
## 1  15:00:00     S-193       T-28       T-44             4             1
## 2  15:00:00     S-192       T-80       T-06             3             0
## 3  12:45:00     S-192       T-84       T-09             2             1
## 4  14:50:00     S-193       T-59       T-54             3             1
## 5  16:00:00     S-192       T-03       T-28             1             0
## 6  14:45:00     S-192       T-13       T-44             3             0
##   ExtraTime PenaltyShootout HomePenalty AwayPenalty        Result
## 1         0               0           ?           0 home team win
## 2         0               0           0           0 home team win
## 3         0               0           0           0 home team win
## 4         0               0           0           ? home team win
## 5         0               0           0           0 home team win
## 6         0               0           0           0 home team win
```

`str()` is used to inspect the structure and variable types, while `head()` presents a small sample of each dataset without printing every observation.

Replace "?" with NA

```r
# Replace every question mark with R's standard missing value.

Tournaments[Tournaments == "?"] <- NA
```

```r
Teams[Teams == "?"] <- NA
Stadiums[Stadiums == "?"] <- NA
Matches[Matches == "?"] <- NA
```

This method searches each data frame and changes all occurrences of `"?"` to `NA`.

Convert categorical variables to factors

```r
# Convert the required categorical variables to factors.

Matches$Result <-as.factor(Matches$Result)
Matches$Stage <- as.factor(Matches$Stage)
Stadiums$Country <- as.factor(Stadiums$Country)
Matches$ExtraTime <- as.factor(Matches$ExtraTime)
```

The actual column name in the dataset is `ExtraTime`. R is case-sensitive, so writing `Extratime` would create or refer to a different variable.

Convert penalty variables to numeric and replace missing values

```r
# Convert the two penalty variables from character to numeric format.

Matches$HomePenalty <- as.numeric(Matches$HomePenalty)
Matches$AwayPenalty <- as.numeric(Matches$AwayPenalty)

# Replace missing penalty values with zero.

Matches$HomePenalty[is.na(Matches$HomePenalty)] <- 0
Matches$AwayPenalty[is.na(Matches$AwayPenalty)] <- 0
```

The missing values are replaced after numeric conversion because the assignment specifically requires missing `HomePenalty` and `AwayPenalty` values to become zero.

Inspect the cleaned data

```r
# Confirm that the required variables now have the correct data types.

str(Matches)
```

```
## 'data.frame':    964 obs. of  16 variables:
##  $ MatchID        : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID   : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName      : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v
##  $ Stage          : Factor w/ 8 levels "final","final round",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ MatchDate      : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime      : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID      : chr  "S-193" "S-192" "S-192" "S-193" ...
```

```
##  $ HomeTeamID     : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID     : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore  : int  4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore  : int  1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ PenaltyShootout: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty    : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ AwayPenalty    : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Result         : Factor w/ 3 levels "away team win",..: 3 3 3 3 3 3 3 3 3 3 ...
```

```
str(Stadiums)
```

```
## 'data.frame':    193 obs. of  4 variables:
##  $ StadiumID  : chr  "S-089" "S-108" "S-107" "S-111" ...
##  $ StadiumName: chr  "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ..
##  $ City       : chr  "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
##  $ Country    : Factor w/ 18 levels "Argentina","Brazil",..: 8 10 10 10 10 6 6 2 2 2 ...
```

```
# Present a summary of the cleaned Matches dataset.

summary(Matches)
```

```
##    MatchID          TournamentID         MatchName        
##  Length:964         Length:964         Length:964        
##  Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character  
##                                                          
##                                                          
##                                                          
##                                                          
##                 Stage      MatchDate          MatchTime        
##  group stage       :676   Length:964         Length:964        
##  round of 16       : 97   Class :character   Class :character  
##  quarter-finals    : 70   Mode  :character   Mode  :character  
##  semi-finals       : 38                                        
##  second group stage: 36                                        
##  final             : 21                                        
##  (Other)           : 26                                        
##   StadiumID          HomeTeamID         AwayTeamID        HomeTeamScore   
##  Length:964         Length:964         Length:964         Min.   : 0.000  
##  Class :character   Class :character   Class :character   1st Qu.: 1.000  
##  Mode  :character   Mode  :character   Mode  :character   Median : 1.000  
##                                                           Mean   : 1.767  
##                                                           3rd Qu.: 3.000  
##                                                           Max.   :10.000  
##                                                                           
##  AwayTeamScore   ExtraTime PenaltyShootout    HomePenalty      AwayPenalty    
##  Min.   :0.000   0:891     Min.   :0.00000   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:0.000   1: 73     1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.0000  
##  Median :1.000             Median :0.00000   Median :0.0000   Median :0.0000  
##  Mean   :1.055             Mean   :0.03631   Mean   :0.1048   Mean   :0.1089  
##  3rd Qu.:2.000             3rd Qu.:0.00000   3rd Qu.:0.0000   3rd Qu.:0.0000  
##  Max.   :7.000             Max.   :1.00000   Max.   :5.0000   Max.   :5.0000  
```

```
##
##             Result
##  away team win:240
##  draw         :179
##  home team win:545
##
##
##
##
```

Identify matches that involved a penalty shootout

```r
# Keep only matches that involved a penalty shootout.
# In the dataset, 1 means that a penalty shootout occurred.

PenaltyMatches <- subset(
  Matches,
  PenaltyShootout == 1
)

# Inspect the filtered data.

head(PenaltyMatches)
```

```
##       MatchID TournamentID                     MatchName          Stage
## 358 M-1982-50      WC-1982         West Germany v France    semi-finals
## 405 M-1986-45      WC-1986              Brazil v France quarter-finals
## 406 M-1986-46      WC-1986         West Germany v Mexico quarter-finals
## 408 M-1986-48      WC-1986              Spain v Belgium quarter-finals
## 453 M-1990-41      WC-1990 Republic of Ireland v Romania    round of 16
## 457 M-1990-45      WC-1990        Argentina v Yugoslavia quarter-finals
##      MatchDate MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore
## 358 07/08/1982  21:00:00     S-159       T-83       T-28             3
## 405  6/21/1986  12:00:00     S-095       T-09       T-28             1
## 406  6/21/1986  16:00:00     S-102       T-83       T-44             0
## 408  6/22/1986  16:00:00     S-104       T-71       T-06             1
## 453  6/25/1990  17:00:00     S-073       T-58       T-59             0
## 457  6/30/1990  17:00:00     S-071       T-03       T-84             0
##     AwayTeamScore ExtraTime PenaltyShootout HomePenalty AwayPenalty
## 358             3         1               1           0           4
## 405             1         1               1           0           4
## 406             0         1               1           0           1
## 408             1         1               1           0           5
## 453             0         1               1           5           4
## 457             0         1               1           3           2
##             Result
## 358 home team win
## 405 away team win
## 406 home team win
## 408 away team win
## 453 home team win
## 457 home team win
```

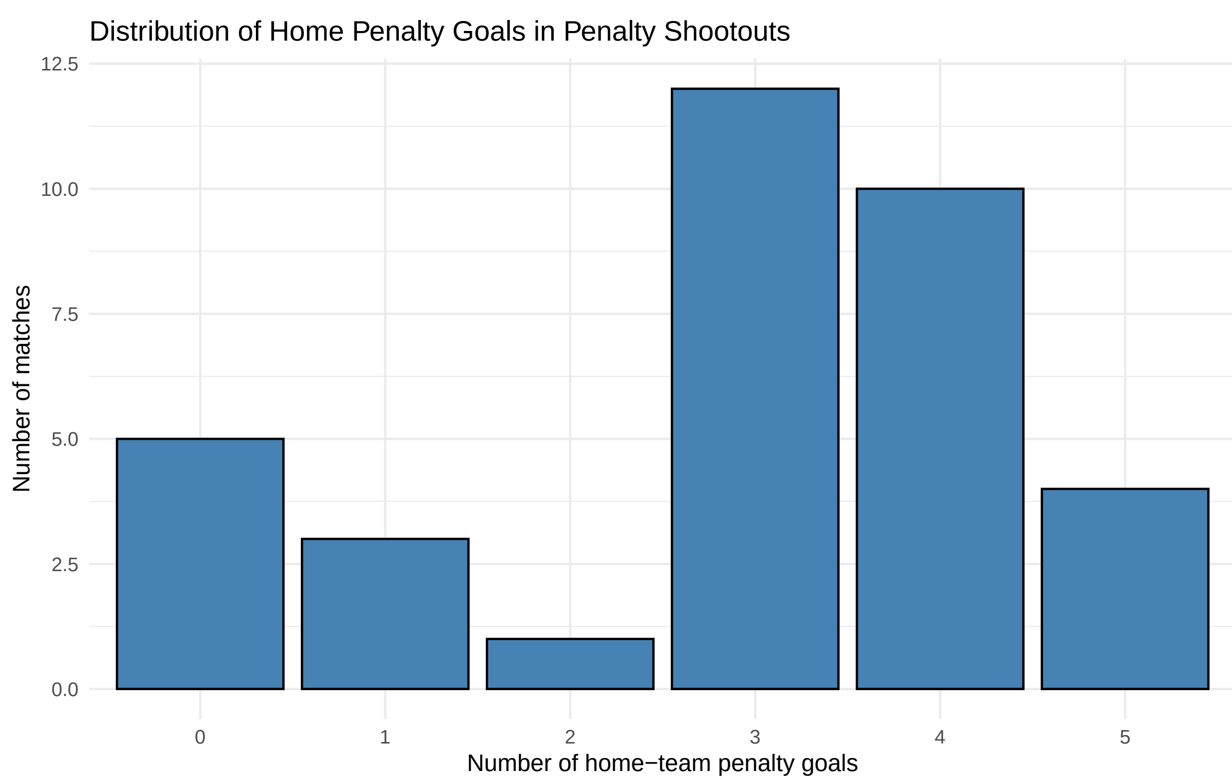
```r
nrow(PenaltyMatches)
```

```
## [1] 35
```

The `subset()` function was used to select only observations where `PenaltyShootout` equals 1. This ensures that the following visualisation does not include ordinary matches that did not proceed to a penalty shootout.

Visualise the distribution of HomePenalty

```r
ggplot(
  data = PenaltyMatches,
  aes(x = factor(HomePenalty))
) +
  geom_bar(
    fill = "steelblue",
    colour = "black"
  ) +
  labs(
    title = "Distribution of Home Penalty Goals in Penalty Shootouts",
    x = "Number of home-team penalty goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Justification of chart choice

Because of being a discrete numerical variable, the bar chart will be the most suitable graph to display 'HomePenalty'. In this, each bar can represent one exact number of home-team penalty goals and the height of the bar shows the frequency of matches.

While 'Matches$HomePenalty' is permanently stored as a numeric value in the dataset, it is temporarily treated as categorical data by writing factor(HomePenalty) in the aes() function. This tells the plotting engine to separate every penalty count without permanently altering the underlying dataframe.

Despite the unsuitable impact of histogram that divides numerical values into intervals, it can meet the demand of these variables that the frequencies of individuals whole-number penalty totals are more meaningful than grouped intervals.

Interpretation

The chart includes 35 matches that were decided by a penalty shootout. Three The most common, occurring in 12 games, is home-team penalty goal. Gave up four goals in 10 games. These values add up to 22 of the 35 observations, with the distribution being primarily centered around three. Four successful penalties, and. Five matches with zero goals, 3 matches with one, 1 match with 2. The five goals he has scored in his last four games, matches. The zero category is valid since a team can shoot but not score points in an original shootout. There are no missing values of 'HomePenalty' in observations. The distribution is skewed to the right with the majority of observations being towards the lower end. There are fewer matches at the other values; near three and four. Overall, three This data set contains the most common home team scores: four penalties. The chart is descriptive but it is not proof that the home-team A specific score of penalty was awarded for designation.

Code testing

```r
# Confirm that the required categorical variables are factors.

stopifnot(is.factor(Matches$Result))
stopifnot(is.factor(Matches$Stage))
stopifnot(is.factor(Stadiums$Country))
stopifnot(is.factor(Matches$ExtraTime))

# Confirm that the penalty variables are numeric.

stopifnot(is.numeric(Matches$HomePenalty))
stopifnot(is.numeric(Matches$AwayPenalty))

# Confirm that the penalty variables have no missing values.

stopifnot(!any(is.na(Matches$HomePenalty)))
stopifnot(!any(is.na(Matches$AwayPenalty)))

# Confirm that the dataset contains penalty-shootout matches.

stopifnot(sum(Matches$PenaltyShootout == 1) == 35)

cat("All Question 1 tests passed successfully.")
```

```
## All Question 1 tests passed successfully.
```

Question 2: Distribution of HomeTeamScore

Rationale

The dataset 'Matches' includes the columns 'HomeTeamScore', 'AwayTeamScore' and 'Stage', while 'TournamentName' is stored in 'Tournaments'. Because of this, the datasets are therefore The tournament data was linked to the players' data by their shared 'TournamentID' so that each match could be analysed. by tournament name. A) they are requested in the task, and B) they are appropriate to use for this type of data. they show how frequently different home-team scores occur. A bin width of one is suitable since the scores in football are whole number counts. The function 'facet_wrap()' will make a histogram for each stage, tournament, and Using away-score group, multiple distributions can be compared in a single analysis. figure. AwayTeamScore is segmented into 0-1, 2-3 and 4+ goal groups with cut(), which does claret and repeatable numeric boundaries. The analysis is concerned with distribution shape, centre, spread and unusually high. To compare scores keeping in mind the difference in the number of matches played in each group.

Combine Matches and Tournaments

```r
# Add TournamentName to the Matches dataset using TournamentID.

Matches2 <- merge(
  Matches,
  Tournaments[, c("TournamentID", "TournamentName")],
  by = "TournamentID",
  all.x = TRUE,
  sort = FALSE
)

# Present the first six rows of the combined dataset.

head(Matches2)
```

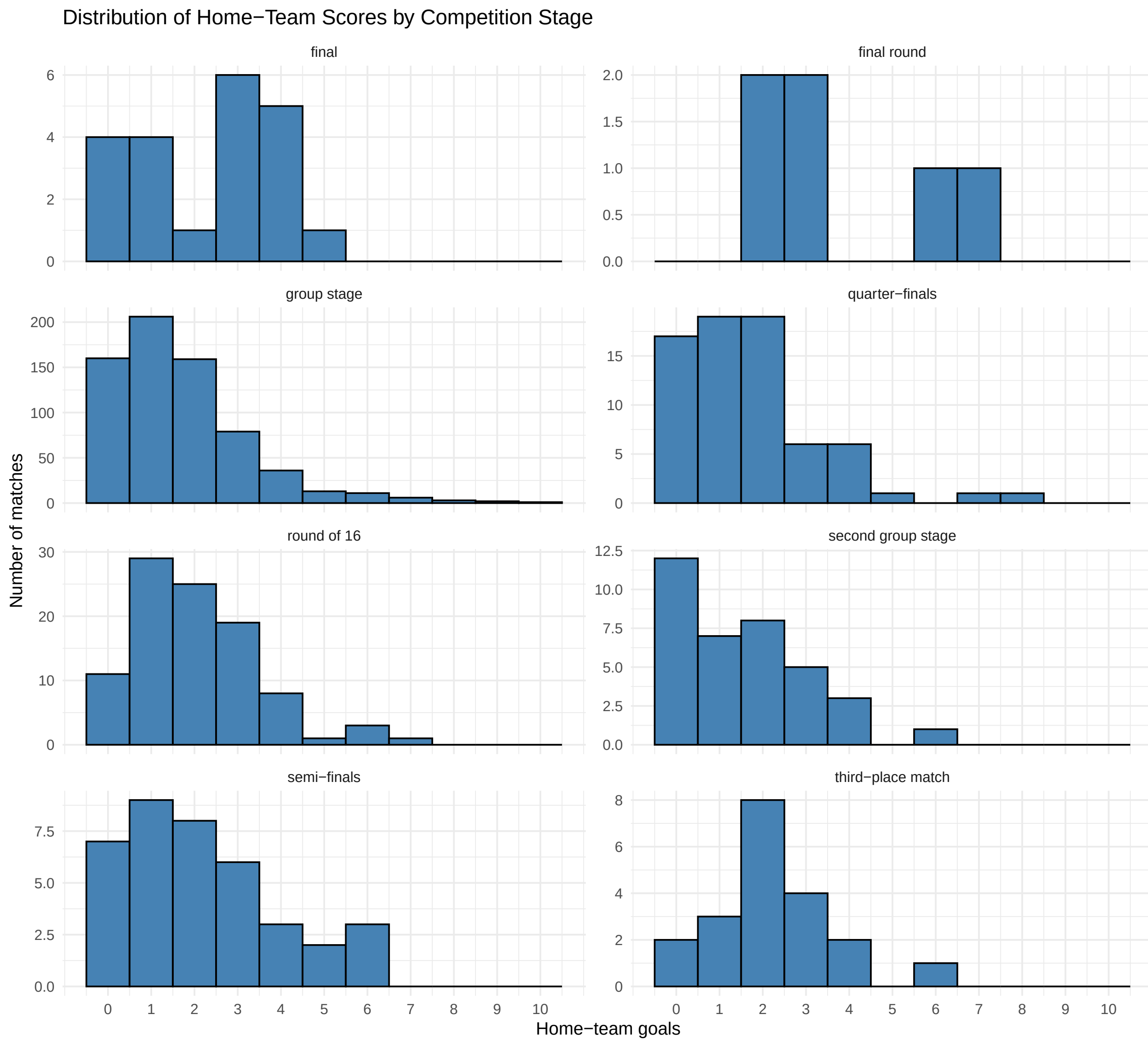
```
##   TournamentID   MatchID               MatchName       Stage MatchDate
## 1      WC-1930 M-1930-01         France v Mexico group stage 7/13/1930
## 2      WC-1930 M-1930-02 United States v Belgium group stage 7/13/1930
## 3      WC-1930 M-1930-03     Yugoslavia v Brazil group stage 7/14/1930
## 4      WC-1930 M-1930-04          Romania v Peru group stage 7/14/1930
## 5      WC-1930 M-1930-05      Argentina v France group stage 7/15/1930
## 6      WC-1930 M-1930-06          Chile v Mexico group stage 7/16/1930
##   MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore AwayTeamScore
## 1  15:00:00     S-193       T-28       T-44             4             1
## 2  15:00:00     S-192       T-80       T-06             3             0
## 3  12:45:00     S-192       T-84       T-09             2             1
## 4  14:50:00     S-193       T-59       T-54             3             1
## 5  16:00:00     S-192       T-03       T-28             1             0
## 6  14:45:00     S-192       T-13       T-44             3             0
##   ExtraTime PenaltyShootout HomePenalty AwayPenalty        Result
## 1         0               0           0           0 home team win
## 2         0               0           0           0 home team win
## 3         0               0           0           0 home team win
## 4         0               0           0           0 home team win
```

```
## 5         0               0           0           0 home team win
## 6         0               0           0           0 home team win
##        TournamentName
## 1 1930 FIFA World Cup
## 2 1930 FIFA World Cup
## 3 1930 FIFA World Cup
## 4 1930 FIFA World Cup
## 5 1930 FIFA World Cup
## 6 1930 FIFA World Cup
```

`all.x = TRUE` keeps all observations from the `Matches` dataset, even if a matching tournament record is not found.

Distribution of HomeTeamScore by Stage

```r
ggplot(
  Matches2,
  aes(x = HomeTeamScore)
) +
  geom_histogram(
    binwidth = 1,
    boundary = -0.5,
    fill = "steelblue",
    colour = "black"
  ) +
  scale_x_continuous(
    breaks = 0:max(Matches2$HomeTeamScore, na.rm = TRUE)
  ) +
  facet_wrap(
    ~ Stage,
    ncol = 2,
    scales = "free_y"
  ) +
  labs(
    title = "Distribution of Home-Team Scores by Competition Stage",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



`scales = "free_y"` allows each panel to use a suitable vertical scale. This makes the distribution visible for stages containing fewer matches. Therefore, the shapes of the distributions can be compared, but the absolute bar heights should be compared carefully because the vertical scales may differ.

Interpretation by Stage

In most competitons, the home team scores fall around the average of zero. and two goals. This stage of the group stage has 676 matches with a median score of one goal, the most common outcome being one goal. There is a round of 16 in the event. A slightly higher centre with a mean of approximately 2.04 and a median of 2. Medians are also two for a semi-final and other third place matches have a median of three goals. The last round is the highest round with the mean of the approximate value of It has a 3.83 rating, but includes just 6 matches. For this reason, there are only a few high-quality quizzes available. Its distribution can be significantly affected by its scores. Several stages have long right tails as occasionally matches contain: High away team scores, even as high as 10 in the group stage. A general trend of lower scores is observed, except for the later knockout In the cases of some stages, higher centres are shown. The differences are to be interpreted are used with great care since the size of samples on the stages varies considerably and the charts do not Consider what factors, like a strong team or old tournament regulations, may have influenced the results.

Distribution of HomeTeamScore by TournamentName

```
ggplot(
  Matches2,
  aes(x = HomeTeamScore)
) +
  geom_histogram(
    binwidth = 1,
    boundary = -0.5,
    fill = "steelblue",
    colour = "black"
  ) +
  scale_x_continuous(
    breaks = 0:max(Matches2$HomeTeamScore, na.rm = TRUE)
  ) +
  facet_wrap(
    ~ TournamentName,
    ncol = 4,
    scales = "free_y"
  ) +
  labs(
    title = "Distribution of Home-Team Scores by FIFA World Cup",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(size = 7),
    axis.text.x = element_text(size = 7)
  )
```


Distribution of Home−Team Scores by FIFA World Cup
1930 FIFA World Cup
1934 FIFA World Cup
1938 FIFA World Cup
1950 FIFA World Cup
1954 FIFA World Cup
1958 FIFA World Cup
1962 FIFA World Cup
1966 FIFA World Cup
1970 FIFA World Cup
1974 FIFA World Cup
1978 FIFA World Cup
1982 FIFA World Cup
1986 FIFA World Cup
1990 FIFA World Cup
1994 FIFA World Cup
1998 FIFA World Cup
2002 FIFA World Cup
2006 FIFA World Cup
2010 FIFA World Cup
2014 FIFA World Cup
2018 FIFA World Cup
2022 FIFA World Cup
Number of matches
Home−team goals

Interpretation by Tournament

The histograms of the tournaments show that there were more World Cups with higher and More dispersed home-team score distributions than previous tournaments. The 1954 FIFA On average, World Cup has the highest amount of goals scored by the home team, around 4.19 goals, followed by 1938 at 3.39 and 1930 at 3.28. These tournaments are also included in the games several matches where the home team scored four or more goals. That is contrasted to modern tournaments, which tend to focus on the 0-1 area, and two goals. The mean was approximately 1.23 in 1990, 1.19 in 2010, and 1.27 in 2014. Some recent competitions still have a few high-scoring isolated competitions. In matches, these observations are not the, but are instead part of long right tails. typical score. As a whole, the pattern of scores indicates that home teams started to score less. More focussed through time. However, the analysis doesn't provide a cause as to why this occurred. The level of the tournament, the type of competition, the quality of the teams, the way the game is played and the defensive tactics have evolved over time, and the name of the tournament (TournamentName) has changed over time. represents multiple differences and not just one cause of that difference.

Create AwayTeamScore groups

```r
# Divide AwayTeamScore into the three groups required by the task.

Matches2$AwayScoreGroup <- cut(
  Matches2$AwayTeamScore,
  breaks = c(-1, 1, 3, Inf),
  labels = c(
    "0-1 goals",
    "2-3 goals",
    "4 or more goals"
  ),
  include.lowest = TRUE
)

# Check the new variable.

str(Matches2$AwayScoreGroup)
```

```
##  Factor w/ 3 levels "0-1 goals","2-3 goals",..: 1 1 1 1 1 1 1 1 1 1 ...
```

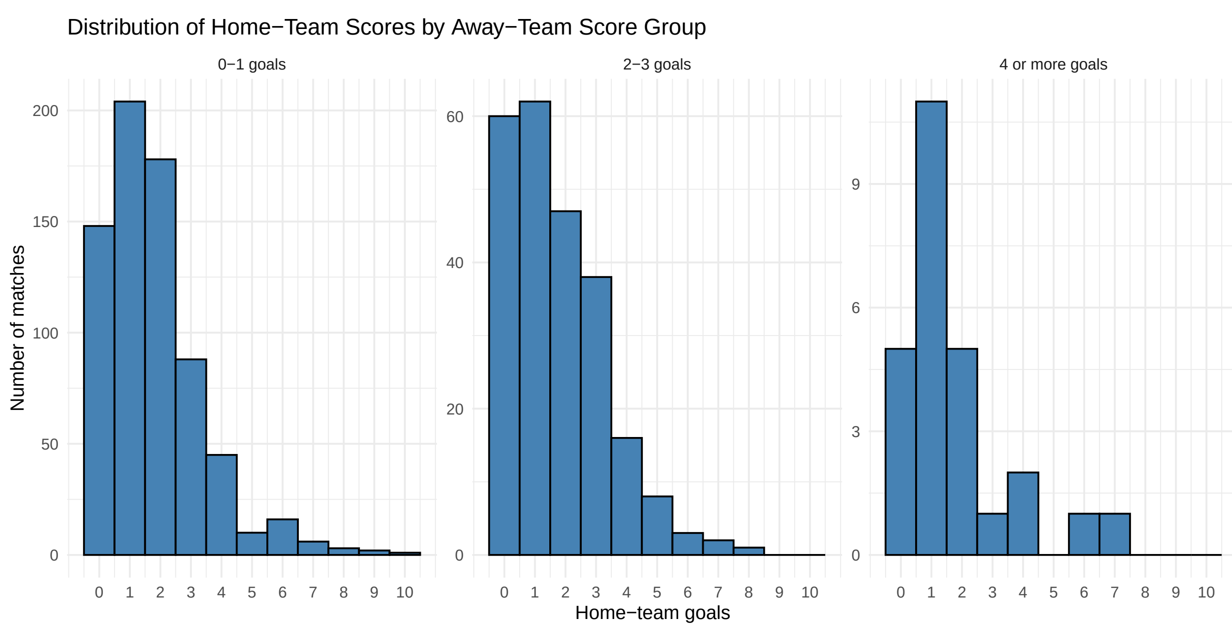
The boundaries create the following groups:

- values from 0 to 1 become `"0-1 goals"`;
- values from 2 to 3 become `"2-3 goals"`;
- values of 4 or more become `"4 or more goals"`.

Distribution of HomeTeamScore by AwayTeamScore group

```r
ggplot(
  Matches2,
  aes(x = HomeTeamScore)
) +
  geom_histogram(
```

```r
    binwidth = 1,
    boundary = -0.5,
    fill = "steelblue",
    colour = "black"
  ) +
  scale_x_continuous(
    breaks = 0:max(Matches2$HomeTeamScore, na.rm = TRUE)
  ) +
  facet_wrap(
    ~ AwayScoreGroup,
    nrow = 1,
    scales = "free_y"
  ) +
  labs(
    title = "Distribution of Home-Team Scores by Away-Team Score Group",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Interpretation by AwayTeamScore group

Away-Team Score Group interpretation is available.Away-Team Score Group interpretation is available. The score of the 3 'AwayTeamScore' groups is fairly similar at home. distributions. In each group, the goal scored by the home team is the most common value. This is also true of the median of the data. The mean HomeTeamScore is ~1.77 when the away team scored. Any goal scored by the ball was equal to 1.75 when the ball scored two goals, and 1.73 when the ball scored three goals. scored 4 or more. The similar averages suggest that the centre of the This is because there are only minor differences in home-team distribution between the groups. The biggest difference is the number of samples. The 0-1 group have 701 matches, 237 are in the 2–3 group and there are only 26 in the 4-or-more group. Consequently, The last one is less smooth, and more susceptible to match-ups. Overall the charts do not offer much visual clue of a significant change in usual. The value of 'HomeTeamScore' gets higher as the value of 'AwayTeamScore' also gets higher. This is a description, a comparison, and the smallest highest scoring group restricts the force of any conclusion

Code testing

```r
# The merge should preserve every row from Matches.

stopifnot(nrow(Matches2) == nrow(Matches))

# Every match should receive a TournamentName.

stopifnot(!any(is.na(Matches2$TournamentName)))

# AwayScoreGroup should be a factor created by cut().

stopifnot(is.factor(Matches2$AwayScoreGroup))

# Every match should be assigned to one of the three groups.

stopifnot(!any(is.na(Matches2$AwayScoreGroup)))

# Test the boundaries of each AwayTeamScore group.

stopifnot(
  all(
    Matches2$AwayTeamScore[
      Matches2$AwayScoreGroup == "0-1 goals"
    ] <= 1
  )
)

stopifnot(
  all(
    Matches2$AwayTeamScore[
      Matches2$AwayScoreGroup == "2-3 goals"
    ] >= 2 &
    Matches2$AwayTeamScore[
      Matches2$AwayScoreGroup == "2-3 goals"
    ] <= 3
  )
)

stopifnot(
  all(
    Matches2$AwayTeamScore[
      Matches2$AwayScoreGroup == "4 or more goals"
    ] >= 4
  )
)

cat("All Question 2 tests passed successfully.")
```

```
## All Question 2 tests passed successfully.
```

If all tests are correct, the final output will display:

```
All Question 2 tests passed successfully.
```

#Question 3: National Teams in Penalty Shootouts

Rationale

Question 3 first uses 'subset()' to select only those matches that 'PenaltyShootout' equals one. Home and Away teams are counted. Are stored separately (HomeTeamID/AwayTeamID) and There are two rankings for task. The function table() is used to see the frequency of each TeamID value and as.data.frame() converts it to a dataframe. Convert the counts into a structure which can be sorted and joined. The counts are Merged to 'Teams' using 'TeamID' as the key as there are no unique keys to merge on in 'Teams' and therefore 'TeamName' and 'TeamCode' are not The data is stored in the 'Matches' data set. 'order()' is a function which sorts the teams from the best to the worst The page with the numbers 1, 2, 3, 4, and 5 will be returned by the query NumberOfMatches, and by head(. . . , 5). The final The only difference between the tables and the table models is that the tables only contain the 'TeamName', 'TeamCode', and 'NumberOfMatches' properties, as do the table models. required output. When using alphabetical ordering, it can be used as a constant rule when The number of participants in each team is the same.

##Filter out the matches decided by a penalty shootout

```
Q3penalty_matches <- subset(
  Matches,
  PenaltyShootout == 1
)
#Check the number of penalty-shootour matches
nrow(Q3penalty_matches)
```

```
## [1] 35
```

```
#Check the accuracy of filtered data
table(Q3penalty_matches$PenaltyShootout)
```

```
##
##  1
## 35
```

##Top 5 home-teams that have most frequently participated in these matches

```
# Count how many penalty shootouts each home team participated in
HomeTeamCounts <- as.data.frame(
  table(Q3penalty_matches$HomeTeamID)
)

# Rename the columns
names(HomeTeamCounts) <- c(
  "TeamID",
  "NumberOfMatches"
```

```
)

# Make sure TeamID has the same data type in both datasets
HomeTeamCounts$TeamID <- as.character(HomeTeamCounts$TeamID)
Teams$TeamID <- as.character(Teams$TeamID)

# Add TeamName and TeamCode using TeamID
HomeTeamResults <- merge(
  HomeTeamCounts,
  Teams,
  by = "TeamID"
)

# Sort from highest to lowest
HomeTeamResults <- HomeTeamResults[
  order(-HomeTeamResults$NumberOfMatches),
]

# Keep the required columns
HomeTeamResults <- HomeTeamResults[
  ,
  c("TeamName", "TeamCode", "NumberOfMatches")
]

# Select the top five away teams
Top5HomeTeams <- head(HomeTeamResults, 5)

Top5HomeTeams
```

```
##        TeamName TeamCode NumberOfMatches
## 1     Argentina      ARG               4
## 2        Brazil      BRA               4
## 17        Spain      ESP               4
## 12  Netherlands      NLD               3
## 20 West Germany      DEU               3
```

##Top 5 away-teams that have most frequently participated in these matches

```
AwayTeamCounts <- as.data.frame(
  table(Q3penalty_matches$AwayTeamID)
)

names(AwayTeamCounts) <- c(
  "TeamID",
  "NumberOfMatches"
)

AwayTeamCounts$TeamID <- as.character(AwayTeamCounts$TeamID)

AwayTeamResults <- merge(
  AwayTeamCounts,
  Teams,
  by = "TeamID"
```

```
)

AwayTeamResults <- AwayTeamResults[
  order(-AwayTeamResults$NumberOfMatches),
]

AwayTeamResults <- AwayTeamResults[
  ,
  c("TeamName", "TeamCode", "NumberOfMatches")
]
#Select the top five away-teams
Top5AwayTeams <- head(AwayTeamResults, 5)

Top5AwayTeams
```

```
##     TeamName TeamCode NumberOfMatches
## 10    France      FRA               5
## 1  Argentina      ARG               3
## 9    England      ENG               3
## 7    Croatia      HRV               2
## 13     Italy      ITA               2
```

Discussion

There is a considerable difference between the ranking of the home team and the ranking of the away team. Argentina, Brazil, Home teams shared the lead with four penalty shootout with Spain. appearances each. 348, followed by the Netherlands and West Germany with three.Three after that, followed by the Netherlands and West Germany, with 348. appearances. Fourth-placed France has five appearances, followed by Italy and Switzerland. There are three in Argentina and three in England. Italy and Croatia fill the away half.The away half is rounded off by Croatia and Italy. 5 players with 2 starts apiece. Argentina is the only team that is in both top-5s, meaning: Teams' participation, whether at home or away, was high, particularly when not recorded. away side. France are the number one away team but not in the home top. Brazil and Spain have the opposite pattern as shown by the numbers five. These are participation-based ranks and not success-based ranks. They aren't a way to show how A number of shootouts that each team had won. They may not be used as evidence of any kind either. There are not a lot of home and away labels to be identified, except the labels World Cup home and World Cup away are mostly used to indicate of home ground advantage. In a match, the two teams that are pitted against each other. In general, the two roles are separated and the Ranking and shows trends that otherwise wouldn't be observed in a cumulative total.

Code testing

```
# Confirm that every filtered observation is a penalty-shootout match.

stopifnot(
  all(Q3penalty_matches$PenaltyShootout == 1)
)

# Confirm that the filtered dataset is not empty.

stopifnot(
```

```r
  nrow(Q3penalty_matches) > 0
)

# Every shootout match must contribute one home-team observation.

stopifnot(
  sum(HomeTeamCounts$NumberOfMatches) ==
    nrow(Q3penalty_matches)
)

# Every shootout match must contribute one away-team observation.

stopifnot(
  sum(AwayTeamCounts$NumberOfMatches) ==
    nrow(Q3penalty_matches)
)

# Confirm that both final tables contain exactly five teams.

stopifnot(
  nrow(Top5HomeTeams) == 5
)

stopifnot(
  nrow(Top5AwayTeams) == 5
)

# Confirm that the final tables contain the required columns.

stopifnot(
  identical(
    names(Top5HomeTeams),
    c(
      "TeamName",
      "TeamCode",
      "NumberOfMatches"
    )
  )
)

stopifnot(
  identical(
    names(Top5AwayTeams),
    c(
      "TeamName",
      "TeamCode",
      "NumberOfMatches"
    )
  )
)

cat("All Question 3 tests passed successfully.")
```

```
## All Question 3 tests passed successfully.
```

Question 4: Factors Associated with Match Outcomes

Rationale

Stage and ExtraTime are categorical factors so these have been selected. available directly in 'Matches' and are more likely to be related to how a match is resolved. The function table() produces cross tabulations of the data between each factor and Result. Raw Percentages are more useful than counts, for checking sample sizes. few players.Too few players in stages and extra time groups to be suitable for comparison. These are the number of matches, which are calculated by the function prop.table(. . . , margin = 1) The percentage for each factor category for the outcome. One-hundred-percent stacked bar charts are used to show the Relative proportions of home team wins, away team wins and draws, and ignoring all groups with the same height, using a different order in the second coordinate. 'coord_flip()' makes all groups easier to read by changing their order in the second coordinate. long stage labels. The analysis is descriptive in nature, and not explanatory. The interpretation also takes into account the different sample sizes and the assumption that the variables 'Stage' and 'ExtraTime' are related to each other.

Overall match outcomes

```r
# Count the total number of matches for each result.

OverallResult <- table(Matches$Result)

OverallResult
```

```
##
## away team win          draw home team win
##           240           179           545
```

There are 545 home-team wins, 240 away-team wins, and 179 draws. Home-team wins are therefore the most common outcome in the complete dataset.

Factor 1: Competition Stage

Create a table of outcomes by Stage

```r
# Count each match outcome within every competition stage.

StageResultCount <- table(
  Matches$Stage,
  Matches$Result
)

# Convert the counts into row percentages.
# margin = 1 calculates percentages within each stage.

StageResultPercent <- round(
```

```r
  prop.table(
    StageResultCount,
    margin = 1
  ) * 100,
  1
)

# Present the percentage table.

knitr::kable(
  StageResultPercent,
  caption = "Percentage of Match Outcomes by Competition Stage"
)
```

Table 1: Percentage of Match Outcomes by Competition Stage

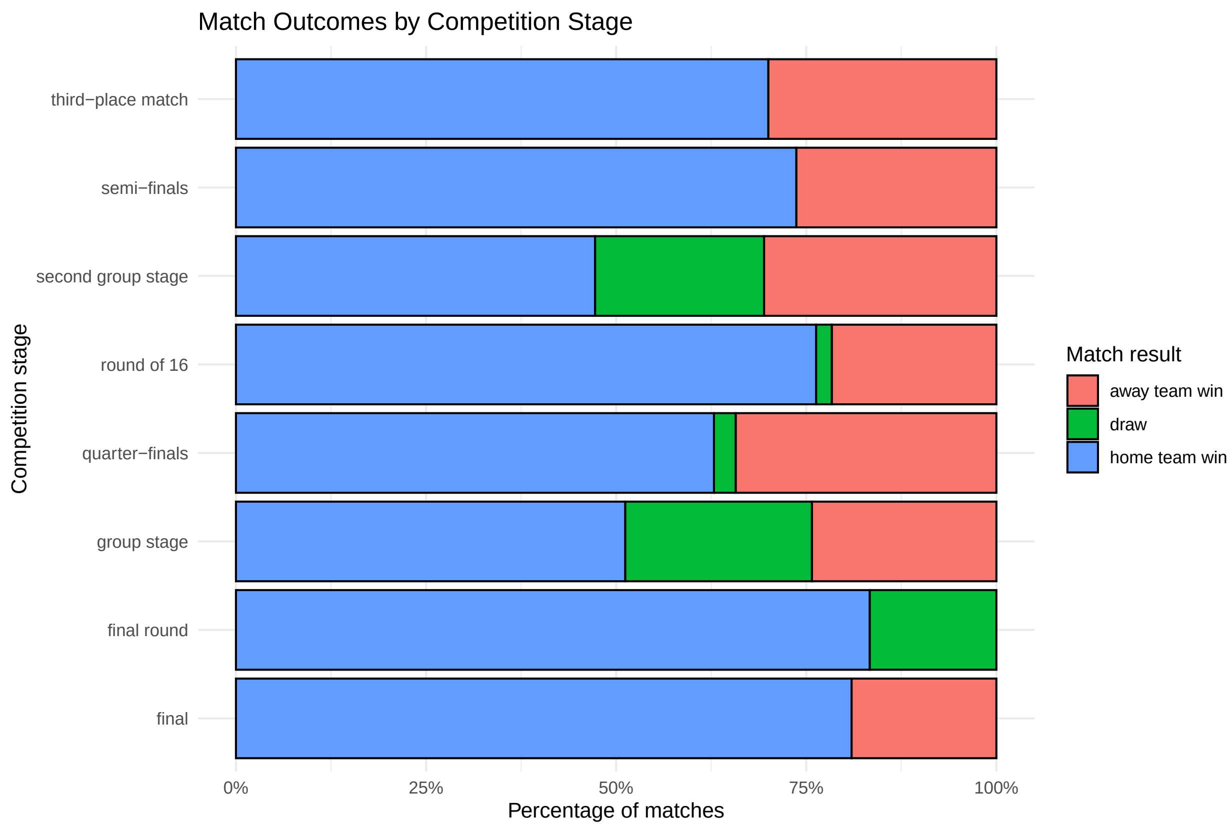
	away team win	draw	home team win
final	19.0	0.0	81.0
final round	0.0	16.7	83.3
group stage	24.3	24.6	51.2
quarter-finals	34.3	2.9	62.9
round of 16	21.6	2.1	76.3
second group stage	30.6	22.2	47.2
semi-finals	26.3	0.0	73.7
third-place match	30.0	0.0	70.0

Each row represents one competition stage and adds to approximately 100%.

Visualise Result by Stage

```r
ggplot(
  Matches,
  aes(
    x = Stage,
    fill = Result
  )
) +
  geom_bar(
    position = "fill",
    colour = "black"
  ) +
  coord_flip() +
  scale_y_continuous(
    labels = scales::percent_format()
  ) +
  labs(
    title = "Match Outcomes by Competition Stage",
    x = "Competition stage",
    y = "Percentage of matches",
    fill = "Match result"
```

```r
) +
theme_minimal()
```



Interpretation of Stage

There is a clear association between 'Result' and competition stage. During the group stage, 51.2% of games are won by the home side, 24.3% are won by the away side. That results in a relatively even distribution of wins and draws with 24.6% of them being draws. distribution. Drops become less frequent in the knockout stages. They make up just 2.1% of round of 16 games and 2.9% of quarter finals, and no games in semi-finals, and Third place matches, or finals. This mirrors the knockout rules of which are: Typically have a winner. 76.3% of round of 16 games, 73.7% of semi-finals and 88.8% of finals are won by home teams.The home team wins 76.3% of round of 16 games, 73.7% of semi-finals and 88.8% of finals. and 81.0% of finals. These percentages are based on a much smaller, however. The 676 group stage games were won by just as many sides as the 125 samples. The data set includes, for instance, only 21 finals. The second group stage is a more even one with 47.2% home victories and 30.6% away victories, and 22.2% draws. Overall, the strongest association of 'Stage' is with the frequency This could be due to the tournament format or team difference, but it's likely due to draws. Extracurricular activities based on the information contained within the text.

Factor 2: Extra Time

Create a table of outcomes by ExtraTime

```r
# Count each match outcome according to whether extra time occurred.
```

```r
ExtraTimeResultCount <- table(
  Matches$ExtraTime,
  Matches$Result
)

# Convert the counts into percentages within each ExtraTime group.

ExtraTimeResultPercent <- round(
  prop.table(
    ExtraTimeResultCount,
    margin = 1
  ) * 100,
  1
)

# Present the percentage table.

knitr::kable(
  ExtraTimeResultPercent,
  caption = "Percentage of Match Outcomes by Extra-Time Status"
)
```

Table 2: Percentage of Match Outcomes by Extra-Time Status

	away team win	draw	home team win
0	24.1	19.4	56.5
1	34.2	8.2	57.5

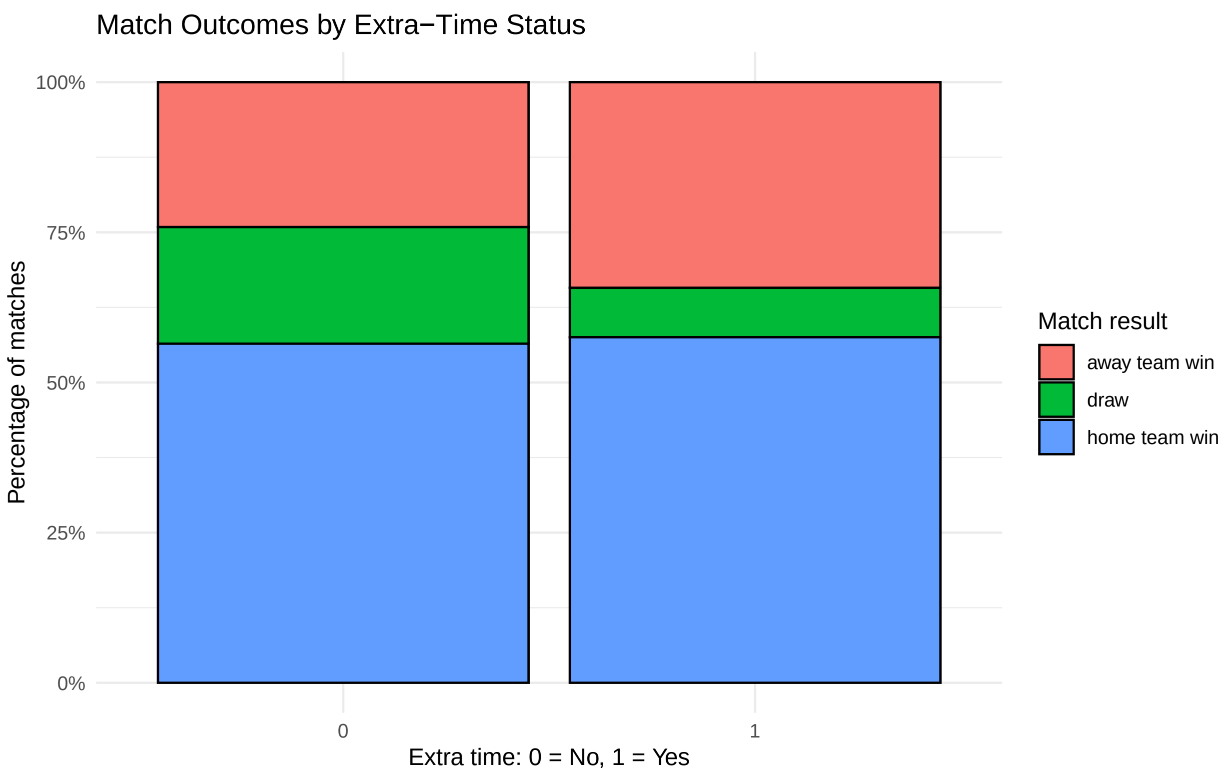
In this variable:

```
0 = no extra time
1 = extra time occurred
```

Visualise Result by ExtraTime

```r
ggplot(
  Matches,
  aes(
    x = ExtraTime,
    fill = Result
  )
) +
  geom_bar(
    position = "fill",
    colour = "black"
  ) +
  scale_y_continuous(
    labels = scales::percent_format()
  ) +
  labs(
```

```
    title = "Match Outcomes by Extra-Time Status",
    x = "Extra time: 0 = No, 1 = Yes",
    y = "Percentage of matches",
    fill = "Match result"
  ) +
  theme_minimal()
```



Interpretation of ExtraTime

The results of the matches also vary based on the 'ExtraTime' property. Among 891 matches without 24.1% in an away-team win, while in extra time, 56.5% of the games were won by the home team. 19.4% in a draw. In 73 extra-time games, 57.5% had been won by the home team, 34.2% had been won by the away team. 8.2% of the matches ended in a draw. However, the proportions for the home-wins are nearly the same, Extra time on the scoreboard makes away-team victories about ten percentage points more likely. Draws are approximately eleven per cent less common, and occurs when it doesn't. The lower draw proportion is understandable, due to the fact that additional time usually occurs. In the knockout rounds if there is a winner required. Some matches might still contain a 'Result' It was a draw and the advancing side was chosen by a penalty shootout at the end of the match. team. As the additional group is quite small, percentage in additional group are more. Has been receptive to individual observations. The pattern thus indicates a association does not lead to any direct change of which side is playing, but just that they are playing together. wins.

Overall discussion

Both of the two factors that have been selected, the factor 'Stage' produces the clearer and more The variations in the 'Result' field are interpretable and, in particular, the difference between the 'group' and 'group2' values is interpretable. matches, with or without draw, and knockout matches. uncommon. A few variables are associated with ExtraTime, such as less drawing and more win away from home. The home-win proportion is stable, and the proportion of home wins drops. The factors are not independent as extra time is predominantly at the end, in the knockout phase. Therefore, An extra time part of the 'ExtraTime' pattern might not be due to additional time, but due to stage rules itself. Other factors which are not measured, such as team quality, opponent power, tournament, etc, also play a role. The results, 'Result', may also be influenced by year, and competition format. The analysis should should therefore be displayed as patterns and relationships within the data set, not as an evidence for a specific result of a match.

Code testing

```r
# Check that Result contains three categories.

stopifnot(
  nlevels(Matches$Result) == 3
)

# Check that Stage and ExtraTime are factors.

stopifnot(
  is.factor(Matches$Stage)
)

stopifnot(
  is.factor(Matches$ExtraTime)
)

# Check that every match was included in the Stage table.

stopifnot(
  sum(StageResultCount) == nrow(Matches)
)

# Check that every match was included in the ExtraTime table.

stopifnot(
  sum(ExtraTimeResultCount) == nrow(Matches)
)

# Check that the percentage-table rows total approximately 100%.

stopifnot(
  all(
    abs(
      rowSums(StageResultPercent) - 100
    ) <= 0.2
  )
)

stopifnot(
  all(
    abs(
```

```r
      rowSums(ExtraTimeResultPercent) - 100
    ) <= 0.2
  )
)

cat("All Question 4 tests passed successfully.")
```

```
## All Question 4 tests passed successfully.
```

#Question 5: GitHub repository and commit on this platform

GitHub Repository

The complete source code, datasets, report files, and development history for this assignment are available in the following GitHub repository:

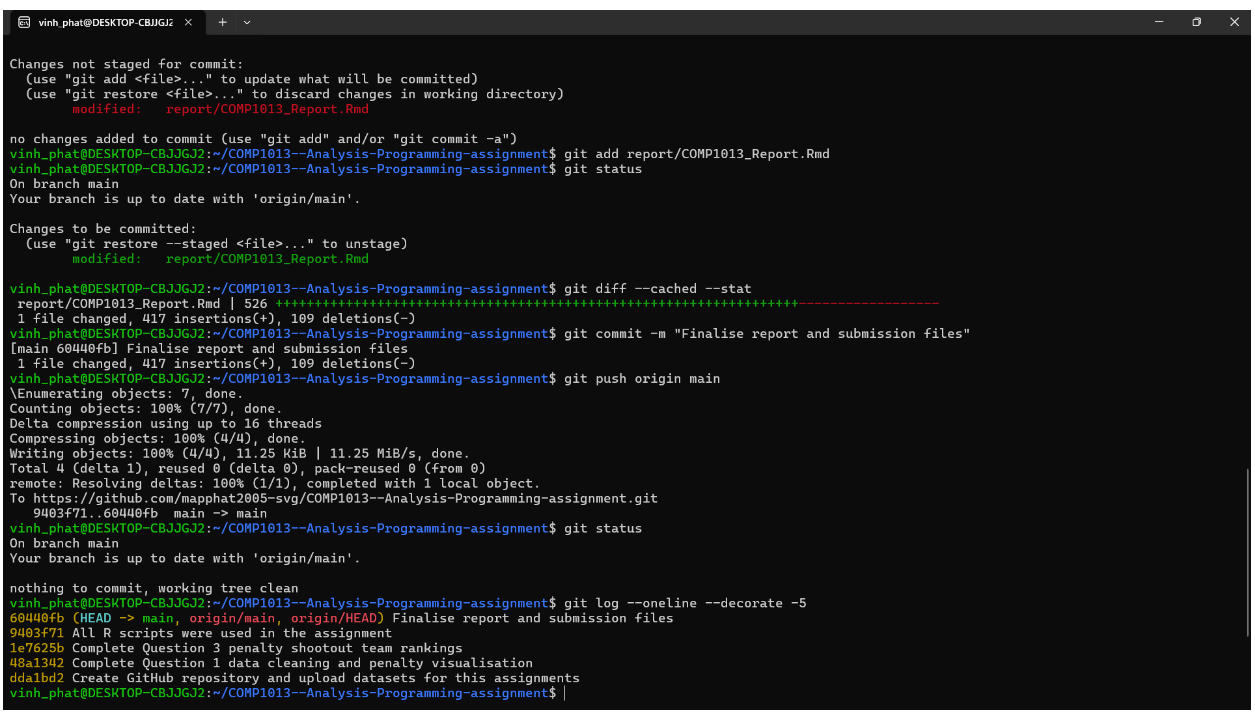
https://github.com/mapphat2005-svg/COMP1013--Analysis-Programming-assignment

Repository Structure

- data/ contains all task dataset
- scripts/ contains the R scripts used to analyze the question 1 to 4
- report/ contains the official report
- figures/ contains the image used in the report

Git Commit History

The following screenshot shows the Git commit history created during the development of the assignment.



```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   report/COMP1013_Report.Rmd

no changes added to commit (use "git add" and/or "git commit -a")
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git add report/COMP1013_Report.Rmd
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   report/COMP1013_Report.Rmd

vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git diff --cached --stat
 report/COMP1013_Report.Rmd | 526 +++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++------------------
 1 file changed, 417 insertions(+), 109 deletions(-)
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git commit -m "Finalise report and submission files"
[main 60440fb] Finalise report and submission files
 1 file changed, 417 insertions(+), 109 deletions(-)
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git push origin main
\Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 11.25 KiB | 11.25 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mapphat2005-svg/COMP1013--Analysis-Programming-assignment.git
   9403f71..60440fb  main -> main
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$ git log --oneline --decorate -5
60440fb (HEAD -> main, origin/main, origin/HEAD) Finalise report and submission files
9403f71 All R scripts were used in the assignment
1e7625b Complete Question 3 penalty shootout team rankings
48a1342 Complete Question 1 data cleaning and penalty visualisation
dda1bd2 Create GitHub repository and upload datasets for this assignments
vinh_phat@DESKTOP-CBJJGJ2:~/COMP1013--Analysis-Programming-assignment$
```

Explanation of Development Commits

The assignment was developed through a series of separate commits rather than being uploaded as one final version. Each commit represents a meaningful stage of development.

1. **Initial commit**
   This commit created the initial GitHub repository and established the main branch.
2. **Create GitHub repository and upload datasets for this assignment**
   This commit added the four original CSV datasets and created the folders for data, scripts, report files, figures, and tables.
3. **Complete Question 1 data cleaning and penalty visualisation**
   This commit added the code for inspecting the datasets, replacing question marks with missing values, converting variable types, processing penalty values, and visualising the distribution of `HomePenalty`.
4. **Complete Question 3 penalty shootout team rankings**
   This commit added the analysis that identified the five home teams and five away teams most frequently involved in penalty shootouts.
5. **All R scripts were used in the assignment**
   This commit reorganised the code from the different questions into one complete R script. The consolidation made it possible to run the assignment analysis from a single file.
6. **Finalise report and submission files**
   This final commit added the completed PDF report, the Git commit-history screenshot, and the final formatting and proofreading changes.